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Information processing apparatus, information processing method, and program

Abstract

An information processing apparatus is an information processing apparatus including a processor, the processor is configured to acquire first image data, output the first image data to a display, acquire visual line data, detect, as a gaze position, a visual line position of a case where a state where a variation distance of the visual line position with respect to a first image indicated by the first image data is equal to or shorter than a first distance continues for a first time or longer based on the visual line data, and set a first region based on two or more gaze positions.

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Background/Summary

(1) This application is a continuation application of International Application No. PCT/JP2022/019585, filed May 6, 2022, the disclosure of which is incorporated herein by reference in its entirety. Further, this application claims priority under 35 USC 119 from Japanese Patent Application No. 2021-161793 filed Sep. 30, 2021, the disclosure of which is incorporated by reference herein.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The technology of the present disclosure relates to an information processing apparatus, an information processing method, and a program.

2. Description of the Related Art

(2) JP1995-199046A (JP-H07-199046A) discloses an apparatus with a visual line detection function comprising a visual line detection unit configured to detect a user's visual line, and a controller configured to detect the user's gazing point from output of the visual line detection unit and determine a region for acquiring information for controlling operations of the apparatus from among a plurality of regions in an observation screen, in which the controller is provided with a varying unit that varies movement responsiveness of the region in a case of determining the region depending on a position in which the user's visual line position exists.

(3) JP2021-105694A discloses an imaging apparatus having a first detection unit configured to detect a position of a gazing point in an image based on the user's visual line, a first setting unit configured to set a range for detecting a subject based on the position of the gazing point and a imaging condition, a second detection unit configured to detect a feature region from the image, and the second setting unit configured to set a focus detection region according to a position of the feature region included in the range for detecting the subject.

(4) JP2000-75198A discloses an apparatus with a visual line detection function that has a plurality of information detection regions and a plurality of visual line detection regions including each of the information detection regions in an image, and performs a predetermined operation based on a signal obtained in an information detection region included in a visual line detection region selected by a visual line detection unit from among the plurality of visual line detection regions, the apparatus having a region calculation unit configured to obtain gaze position-valid regions of an operator in an image, calculate a gaze overlapping region in which all of the plurality of gaze position-valid regions overlap each other, and select a region, which is included in a visual line detection region including the gaze overlapping region, as the information detection region in each

case where the visual line detection unit performs visual line detection.

SUMMARY OF THE INVENTION

- (5) One embodiment according to the technology of the present disclosure provides an information processing apparatus, an information processing method, and a program capable of setting a region on an image based on a visual line position, for example.
- (6) A first aspect according to the technology of the present disclosure is an information processing apparatus comprising a processor, in which the processor is configured to acquire first image data, output the first image data to a display, acquire visual line data, detect, as a gaze position, a visual line position of a case where a state where a variation distance of the visual line position with respect to a first image indicated by the first image data is equal to or shorter than a first distance continues for a first time or longer based on the visual line data, and set a first region based on two or more gaze positions.
- (7) A second aspect according to the technology of the present disclosure is the information processing apparatus according to the first aspect, in which the processor is configured to set, as the first region, a rectangular region having two gaze positions as opposite corners, a circular region having a line connecting the two gaze positions as a diameter, or a closed region defined by connecting a plurality of gaze positions in an order in which the plurality of gaze positions are detected.
- (8) A third aspect according to the technology of the present disclosure is the information processing apparatus according to the first or second aspect, in which the processor is configured to, in a case where three or more gaze positions are detected, set a position at which lines connecting two gaze positions intersect with each other as an intersection position, and set the first region based on the gaze positions and the intersection position.
- (9) A fourth aspect according to the technology of the present disclosure is the information processing apparatus according to the third aspect, in which, in a case where the first region having n vertices is set, at least $n+1$ gaze positions are used.
- (10) A fifth aspect according to the technology of the present disclosure is the information processing apparatus according to any one of the first to fourth aspects, in which the processor is configured to output data indicating a visual line position mark that specifies the visual line position in the first image until the first time is reached, and output data indicating a gaze position mark that specifies the gaze position in the first image in a case where the first time is reached.
- (11) A sixth aspect of the technology of the present disclosure is the information processing apparatus according to the fifth aspect, in which the visual line position mark changes in form with passage of time.
- (12) A seventh aspect according to the technology of the present disclosure is the information processing apparatus according to any one of the first to sixth aspects, in which the processor is configured to set the first region from a plurality of regions included in the first image based on the two or more gaze positions.
- (13) An eighth aspect of the technology of the present disclosure is the information processing apparatus according to any one of the first to seventh aspects, in which the information processing apparatus is an imaging apparatus, and the processor is configured to perform, in a case where a target subject is included as an image in the first region, control to bring the target subject in a depth of field.
- (14) “A ninth aspect according to the technology of the present disclosure is an information processing apparatus comprising a processor, in which the processor is configured to acquire second image data, output the second image data to a display, acquire visual line data, detect a track along which a visual line position has varied within a second time with respect to a second image indicated by the second image data based on the visual line data, and set a second region based on the track.”
- (15) A tenth aspect according to the technology of the present disclosure is the information

processing apparatus according to the ninth aspect, in which the second region is a region surrounded by the track.

(16) An eleventh aspect according to the technology of the present disclosure is the information processing apparatus according to the ninth aspect, in which the second region is a region surrounding the track.

(17) A twelfth aspect according to the technology of the present disclosure is the information processing apparatus according to any one of the ninth to eleventh aspects, in which the processor is configured to determine whether or not a blink occurs within the second time based on the visual line data, and detect, in a case of determining that the blink occurs, the track based on a line connecting a first visual line position before the occurrence of the blink and a second visual line position after the occurrence of the blink.

(18) A thirteenth aspect according to the technology of the present disclosure is the information processing apparatus according to any one of the ninth to twelfth aspects, in which the processor is configured to set the second region from a plurality of regions included in the second image based on the track.

(19) A fourteenth aspect of the technology of the present disclosure is the information processing apparatus according to any one of the ninth to thirteenth aspects, in which the information processing apparatus is an imaging apparatus, and the processor is configured to perform, in a case where a target subject is included as an image in the second region, control to bring the target subject within a depth of field.

(20) A fifteenth aspect according to the technology of the present disclosure is an information processing method comprising acquiring first image data, outputting the first image data to a display, acquiring visual line data, detecting, as a gaze position, a visual line position of a case where a state where a variation distance of the visual line position with respect to a first image indicated by the first image data is equal to or shorter than a first distance continues for a first time or longer based on the visual line data, and setting a first region based on two or more gaze positions.

(21) A sixteenth aspect according to the technology of the present disclosure is an information processing method comprising acquiring second image data, outputting the second image data to a display, acquiring visual line data, detecting a track along which a visual line position has varied within a second time with respect to a second image indicated by the second image data based on the visual line data, and setting a second region based on the track.

(22) A seventeenth aspect according to the technology of the present disclosure is a program for causing a computer to execute a process comprising acquiring first image data, outputting the first image data to a display, acquiring visual line data, detecting, as a gaze position, a visual line position of a case where a state where a variation distance of the visual line position with respect to a first image indicated by the first image data is equal to or shorter than a first distance continues for a first time or longer based on the visual line data, and setting a first region based on two or more gaze positions.

(23) An eighteenth aspect according to the technology of the present disclosure is a program for causing a computer to execute a process comprising acquiring second image data, outputting the second image data to a display, acquiring visual line data, detecting a track along which a visual line position has varied within a second time with respect to a second image indicated by the second image data based on the visual line data, and setting a second region based on the track.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a two view diagram showing an example of a side surface and a rear surface of an

imaging apparatus.

(2) FIG. 2 is a block diagram showing an example of a hardware configuration of the imaging apparatus.

(3) FIG. 3 is an explanatory diagram showing an example of a method of detecting a visual line position.

(4) FIG. 4 is a block diagram showing an example of a functional configuration of a processor according to a first embodiment.

(5) FIG. 5 is an explanatory diagram showing an example of a first operation of the processor according to the first embodiment.

(6) FIG. 6 is an explanatory diagram showing an example of a second operation of the processor according to the first embodiment.

(7) FIG. 7 is an explanatory diagram showing an example of a third operation of the processor according to the first embodiment.

(8) FIG. 8 is an explanatory diagram showing an example of a fourth operation of the processor according to the first embodiment.

(9) FIG. 9 is an explanatory diagram showing an example of a fifth operation of the processor according to the first embodiment.

(10) FIG. 10 is a flowchart showing an example of a flow of target region setting processing according to the first embodiment.

(11) FIG. 11 is an explanatory diagram showing a modification example of the operation of the processor according to the first embodiment.

(12) FIG. 12 is an explanatory diagram showing an example of an operation of a processor according to a second embodiment.

(13) FIG. 13 is an explanatory diagram showing an example of a first operation of a processor according to a third embodiment.

(14) FIG. 14 is an explanatory diagram showing an example of a second operation of the processor according to the third embodiment.

(15) FIG. 15 is an explanatory diagram showing an example of a first operation of a processor according to a fourth embodiment.

(16) FIG. 16 is an explanatory diagram showing an example of a second operation of the processor according to the fourth embodiment.

(17) FIG. 17 is a flowchart showing an example of a flow of target region setting processing according to the fourth embodiment.

(18) FIG. 18 is an explanatory diagram showing a first modification example of the operation of the processor according to the fourth embodiment.

(19) FIG. 19 is an explanatory diagram showing a second modification example of the operation of the processor according to the fourth embodiment.

(20) FIG. 20 is an explanatory diagram showing an example of an operation of a processor according to a fifth embodiment.

(21) FIG. 21 is a flowchart showing an example of a flow of target region setting processing according to the fifth embodiment.

(22) FIG. 22 is an explanatory diagram showing an example of a first operation of a processor according to a sixth embodiment.

(23) FIG. 23 is an explanatory diagram showing an example of a second operation of the processor according to the sixth embodiment.

(24) FIG. 24 is an explanatory diagram showing an example of a third operation of the processor according to the sixth embodiment.

(25) FIG. 25 is an explanatory diagram showing an example of a fourth operation of the processor according to the sixth embodiment.

(26) FIG. 26 is a flowchart showing an example of a flow of target region setting processing

according to the sixth embodiment.

(27) FIG. **27** is an explanatory diagram showing an example of a first operation of a processor according to a seventh embodiment.

(28) FIG. **28** is an explanatory diagram showing an example of a second operation of the processor according to the seventh embodiment.

(29) FIG. **29** is a flowchart showing an example of a flow of target region setting processing according to the seventh embodiment.

(30) FIG. **30** is an explanatory diagram showing an example of an operation of a processor according to an eighth embodiment.

(31) FIG. **31** is a flowchart showing an example of a flow of autofocus processing according to the eighth embodiment.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(32) Hereinafter, one example of an embodiment of an information processing apparatus, an information processing method, and a program according to the technology of the present disclosure will be described in accordance with the appended drawings.

(33) First, terms that are used in the following description will be described.

(34) OF is an abbreviation for “Interface”. CPU is an abbreviation for “Central Processing Unit”. NVM is an abbreviation for “Non-Volatile Memory”. RAM is an abbreviation for “Random Access Memory”. EEPROM is an abbreviation for “Electrically Erasable and Programmable Read Only Memory”. HDD is an abbreviation for “Hard Disk Drive”. CMOS is an abbreviation for “Complementary Metal Oxide Semiconductor”. CCD is an abbreviation for “Charge Coupled Device”. SSD is an abbreviation for “Solid State Drive”. EL is an abbreviation for “Electro Luminescence”. LED is an abbreviation for “Light Emitting Diode”. OLED is an abbreviation for “Organic Light-Emitting Diode”. GPU is an abbreviation for “Graphics Processing Unit”. TPU is an abbreviation for “Tensor Processing Unit”. USB is an abbreviation for “Universal Serial Bus”. ASIC is an abbreviation for “Application Specific Integrated Circuit”. FPGA is an abbreviation for “Field-Programmable Gate Array”. PLD is an abbreviation for “Programmable Logic Device”. SoC is an abbreviation for “System-on-a-Chip”. IC is an abbreviation for “Integrated Circuit”.

First Embodiment

(35) First, a first embodiment will be described.

(36) As shown in FIG. **1** as an example, an imaging apparatus **10** is a digital camera and comprises a lens unit **12** and an imaging apparatus main body **14**. The imaging apparatus **10** is an example of an “information processing apparatus” and an “imaging apparatus” according to the technology of the present disclosure. A pitch axis, a yaw axis, and a roll axis are defined in the imaging apparatus **10**. In the example shown in FIG. **1**, an axis P indicates the pitch axis of the imaging apparatus **10**, an axis Y indicates the yaw axis of the imaging apparatus **10**, and an axis R indicates the roll axis of the imaging apparatus **10**.

(37) The lens unit **12** is attached to the imaging apparatus main body **14**. A touch panel display **34** and an instruction key **36** are provided on a rear surface of the imaging apparatus main body **14**. The touch panel display **34** is formed by a touch panel **34A** and a display **34B**, and for example, the touch panel **34A** is superimposed on the display **34B**.

(38) An electronic viewfinder **28** is provided on an upper portion of the imaging apparatus main body **14**. The electronic viewfinder **28** includes a display **72**. A finder opening portion **16** is provided in the upper portion of the imaging apparatus main body **14**, and the display **72** is provided inside the finder opening portion **16**. The display **72** is disposed at a position where a user can see the display **72** with an eye **302** when looking into the finder opening portion **16**. The display **72** is an example of a “display” according to the technology of the present disclosure.

(39) A light source **82** and a visual line sensor **86** for visual line detection are provided inside the finder opening portion **16**. The user's eye **302** looking into the finder opening portion **16** is a visual line detection target. The light source **82** includes a plurality of light emitters **82A**. The plurality of

light emitters **82A** are arranged side by side in a pitch axis direction of the imaging apparatus **10**, for example.

(40) The light source **82** is disposed at a position where the light source **82** can irradiate the user's eye **302** with light in a case where the user looks into the finder opening portion **16**. The light source **82** emits near-infrared light, for example. The visual line sensor **86** is disposed at a position where an image of the user's eye **302** can be captured when the user looks into the finder opening portion **16**.

(41) A half mirror (not shown) may be disposed inside the finder opening portion **16**. The display **72** may be disposed on a passing optical axis of the half mirror, and the visual line sensor **86** may be disposed on the reflection optical axis of the half mirror.

(42) As shown in FIG. 2 as an example, the imaging apparatus main body **14** comprises a computer **20**, an image sensor **24**, an image sensor control circuit **26**, an electronic viewfinder **28**, a visual line detection unit **30**, an image memory **32**, a UI system unit **92**, and an input/output OF **38**. The image sensor **24**, the image sensor control circuit **26**, the image memory **32**, and the UI system unit **92** are connected to the input/output OF **38**.

(43) The computer **20** comprises a processor **42**, a NVM **44**, and a RAM **46**. The processor **42** controls the entire imaging apparatus **10**. The processor **42** is, for example, a processing unit including a CPU and a GPU, and the GPU operates under the control of the CPU and executes processing related to an image. Here, a processing unit including a CPU and a GPU has been described as an example of the processor **42**, but this is merely an example, and the processor **42** may be one or more CPUs into which a GPU function is integrated, or may be one or more CPUs into which a GPU function is not integrated. The processor **42**, the NVM **44**, and the RAM **46** are connected via a bus **48**, and the bus **48** is connected to the input/output I/F **38**. The computer **20** is an example of a "computer" according to the technology of the present disclosure. The processor **42** is an example of a "processor" according to the technology of the present disclosure.

(44) The NVM **44** is a non-transitory storage medium and stores various parameters and various programs. For example, the NVM **44** is a flash memory (for example, EEPROM). However, this is merely an example, and an HDD or the like may be applied as NVM **44** together with a flash memory. A RAM **46** temporarily stores various kinds of information and is used as a work memory.

(45) The processor **42** reads a necessary program from the NVM **44** and executes the read program in the RAM **46**. The processor **42** controls the image sensor **24**, the image sensor control circuit **26**, the electronic viewfinder **28**, the visual line detection unit **30**, the image memory **32**, and the UI system unit **92** according to the program executed in the RAM **46**.

(46) The image sensor **24** is, for example, a CMOS image sensor. Here, although the CMOS image sensor is exemplified as the image sensor **24**, the technology of the present disclosure is not limited thereto, and for example, even in a case where the image sensor **24** is another type of image sensor such as a CCD image sensor, the technology of the present disclosure is valid. An image sensor control circuit **26** is connected to the image sensor **24**. The image sensor control circuit **26** controls the image sensor **24** according to an imaging control signal from the processor **42**.

(47) Subject light is incident on an imaging lens **52** of the lens unit **12** to be described later. The subject light is imaged on a light-receiving surface of the image sensor **24** by the imaging lens **52**. Photoelectric conversion element (not shown) is provided on the light-receiving surface of the image sensor **24**. The photoelectric conversion elements photoelectrically convert the subject light received by the light-receiving surface and output an electric signal corresponding to an amount of the subject light as analog image data indicating the subject light under the control of the image sensor control circuit **26**. The image sensor **24** includes a signal processing circuit (not shown). The signal processing circuit generates digital captured image data by digitizing the analog image data, and outputs the captured image data.

(48) The captured image data generated by the image sensor **24** is temporarily stored in the image memory **32**. The processor **42** acquires the captured image data from the image memory **32** and

executes various kinds of processing using the acquired captured image data.

(49) The lens unit **12** comprises the imaging lens **52**. The imaging lens **52** has, for example, an objective lens **54**, a focus lens **56**, a zoom lens **58**, and a stop **60**. Further, the lens unit **12** comprises a lens control circuit **62**, a first actuator **64**, a second actuator **66**, and a third actuator **68**. The lens control circuit **62** is connected to the first actuator **64**, the second actuator **66**, and the third actuator **68**, and the lens control circuit **62** is connected to the input/output OF **38**. The lens control circuit **62** controls the first actuator **64**, the second actuator **66**, and the third actuator **68** according to a lens control signal from the processor **42**.

(50) The first actuator **64** moves the focus lens **56** along an optical axis OA. A position of focus is adjusted by changing a position of the focus lens **56**. The second actuator **66** moves the zoom lens **58** along the optical axis OA. A focal length is adjusted by changing a position of the zoom lens **58**. The third actuator **68** changes a size of an opening of the stop **60**. A stop amount by the stop **60** is changed by changing the size of the opening of the stop **60** so that an exposure is adjusted. The first actuator **64**, the second actuator **66**, and the third actuator **68** are, for example, piezoelectric elements, voice coil motors, or the like.

(51) The electronic viewfinder **28** includes the display **72**, a display control circuit **74**, and an eyepiece lens **76**. The display **72** is, for example, a liquid crystal display, an EL display, or the like. The display control circuit **74** is connected to the input/output OF **38**. The processor **42** selectively outputs the captured image data, visual line position data, gaze position data, and target region data to the display control circuit **74** as described later. The display control circuit **74** displays an image on the display **72** according to the captured image data, the visual line position data, the gaze position data, and the target region data. The eyepiece lens **76** is disposed to face a screen of the display **72**.

(52) The visual line detection unit **30** has the light source **82**, a light source control circuit **84**, the visual line sensor **86**, and a visual line sensor control circuit **88**. The light source **82** is, for example, an LED that outputs near-infrared light. The light source control circuit **84** is connected to the input/output OF **38**. The light source control circuit **84** controls the light source **82** according to a light source control signal from the processor **42**.

(53) The visual line sensor **86** is, for example, a CMOS image sensor having sensitivity to near-infrared light. Here, although the CMOS image sensor is exemplified as the visual line sensor **86**, the technology of the present disclosure is not limited thereto, and for example, even in a case where the visual line sensor **86** is another type of image sensor such as a CCD image sensor, the technology of the present disclosure is valid. The visual line sensor control circuit **88** is connected to the visual line sensor **86**. The visual line sensor control circuit **88** controls the visual line sensor **86** according to an imaging control signal from the processor **42**. The visual line sensor **86** captures an image of a subject (for example, the user's eye **302**), and outputs visual line data obtained by capturing the image.

(54) The UI system unit **92** comprises the display **34B**. The processor **42** outputs display image data to the display **34B**, and causes the display **34B** to display an image and various kinds of information based on the display image data. Further, the UI system unit **92** comprises a receiving unit **94** that receives an instruction from the user. The receiving unit **94** comprises the touch panel **34A** and a hard key unit **96**. The hard key unit **96** is a plurality of hard keys including the instruction key **36** (refer to FIG. 1). The receiving unit **94** outputs reception data corresponding to various instructions received by the touch panel **34A** and/or the hard key unit **96**.

(55) FIG. 3 shows an example of a method of detecting a visual line position based on the visual line data. In the example shown in FIG. 3, near-infrared light is emitted from the plurality of light emitters **82A** to an eyeball **306**, and the near-infrared light is reflected by a surface of a cornea **308**, whereby a dot pattern **234** is formed on the surface of the cornea **308**. A distance **L1** between the patterns **234** are obtained from an image **238** obtained by imaging the eyeball **306** with the visual line sensor **86**.

(56) A distance L3 between the visual line sensor **86** and the eyeball **306** is obtained from a known distance L2 between the plurality of light emitters **82A**, a known attachment angle α of the plurality of light emitters **82A**, a known attachment angle of the visual line sensor **86**, and the distance L1 between the patterns **234**. Further, positions of a center **310A** of a pupil **310** and an outer shape **310B** of the pupil **310** with respect to a center **238A** of the image **238** are obtained from the image **238** obtained by imaging the eyeball **306** with the visual line sensor **86**. Coordinates of a center **306A** of the eyeball **306** and coordinates of the center **310A** of the pupil **310** are obtained based on the positions of the center **310A** of the pupil **310** and the outer shape **310B** of the pupil **310** with respect to the center **238A** of the image **238**, the distance L3 between the visual line sensor **86** and the eyeball **306**, and a curvature radius predetermined for the eyeball **306**. For example, a statistically obtained numerical value is applied to the curvature radius of the eyeball **306**.

(57) A direction in which a line **312** connecting the coordinates of the center **306A** of the eyeball **306** and the coordinates of the center **310A** of the pupil **310** extends corresponds to a direction of a visual line of the eye **302**. Further, an extension line **314** obtained by extending the line **312** connecting the coordinates of the center **306A** of the eyeball **306** and the coordinates of the center **310A** of the pupil **310** indicates the visual line of the eye **302**, and a point Pat which the visual line of the eye **302** and the screen of the display **72** intersect corresponds to a position of the visual line of the eye **302**. As described above, the position of the visual line of the eye **302** is detected based on the visual line data obtained by imaging the eyeball **306** with the visual line sensor **86**.

Hereinafter, the “position of the visual line of the eye **302**” is also referred to as a “visual line position” or a “visual line position with respect to the display **72**”. The visual line position is an example of a “visual line position” according to the technology of the present disclosure.

(58) Note that the above-described method of detecting the visual line position is merely an example, and the technology of the present disclosure is not limited thereto. In addition to the above-described method of detecting the visual line position, various methods of detecting the visual line position with respect to the display **72** based on the visual line data obtained by imaging the eyeball **306** with the visual line sensor **86** in a state where the patterns **234** are formed on the surface of the cornea **308** can be applied to the technology of the present disclosure. Further, although the visual line position is detected based on reflected light reflected by the surface of the cornea **308** in the above-described method of detecting the visual line position, the visual line position may be detected based on reflected light reflected by a portion other than the cornea **308** (for example, a retina or the like).

(59) As shown in FIG. 4 as an example, a program **100** is stored in the NVM **44** of the imaging apparatus **10**. The program **100** is an example of a “program” according to the technology of the present disclosure. The processor **42** reads the program **100** from the NVM **44** and executes the read program **100** in the RAM **46**. The processor **42** performs target region setting processing according to the program **100** executed in the RAM **46**. The target region setting processing is executed by the processor **42** operating as a captured image display control unit **102**, a visual line detection processing unit **104**, a visual line position display control unit **106**, a gaze position display control unit **108**, and a target region setting unit **110** according to the program **100**.

(60) The imaging apparatus **10** has a plurality of operation modes and operates under a designated operation mode. For example, the target region setting processing is executed in a case where a target region setting mode is set as the operation mode of the imaging apparatus **10**. The target region setting mode is set in the imaging apparatus **10**, for example, in a case where a target region setting mode setting instruction is received by the receiving unit **94** (refer to FIG. 2). The target region setting mode setting instruction refers to, for example, an instruction to shift from an operation mode other than the target region setting mode to the target region setting mode.

(61) As shown in FIG. 5 as an example, the captured image display control unit **102** causes the display **72** to display a captured image **200**. Specifically, the captured image display control unit **102** performs the following processing. That is, the captured image display control unit **102** causes

the image sensor **24** to image the subject (not shown) through the image sensor control circuit **26** by outputting the imaging control signal to the image sensor control circuit **26**. Subsequently, the captured image display control unit **102** acquires the captured image data obtained by imaging the subject with the image sensor **24**, and outputs the acquired captured image data to the display **72**. The display **72** displays the captured image **200** indicated by the captured image data. Accordingly, the captured image **200** in which the subject is included as an image is displayed on the display **72**. The captured image data is an example of “first image data” according to the technology of the present disclosure. The captured image **200** is an example of a “first image” according to the technology of the present disclosure.

(62) As shown in FIG. **6** as an example, the visual line detection processing unit **104** detects the visual line position of the user viewing the display **72** on which the captured image **200** is displayed. Specifically, the visual line detection processing unit **104** performs the following processing. That is, the visual line detection processing unit **104** causes the visual line sensor **86** to image the subject (for example, the user's eye **302**) through the visual line sensor control circuit **88** by outputting the imaging control signal to the visual line sensor control circuit **88**. Then, the visual line detection processing unit **104** acquires the visual line data obtained by being imaged by the visual line sensor **86**, and detects the visual line position with respect to the display **72** based on the acquired visual line data. The method of detecting the visual line position is as described in FIG. **3**. Thus, the visual line position of the user looking at the display **72** on which the captured image **200** is displayed is detected. The visual line data is an example of “visual line data” according to the technology of the present disclosure. The visual line detection processing unit **104** generates the visual line position data indicating the visual line position and stores the visual line position data in the RAM **46** in a case of detecting the visual line position.

(63) The light source **82** may continue to emit the near-infrared light in a case where power of the imaging apparatus **10** is turned on. Further, the light source **82** may emit the near-infrared light in a case where the visual line position is detected by the visual line detection processing unit **104**, and may stop emitting the near-infrared light in a case where the visual line position is not detected by the visual line detection processing unit **104**.

(64) As shown in FIG. **7** as an example, the visual line position display control unit **106** causes the display **72** to display a visual line position mark **202** corresponding to the visual line position. Specifically, the visual line position display control unit **106** performs the following processing. That is, the visual line position display control unit **106** acquires all the visual line position data stored in the RAM **46** since the target region setting processing is started. Then, the visual line position display control unit **106** determines whether or not a variation distance of the visual line position is equal to or shorter than a predetermined distance based on the acquired visual line position data.

(65) The variation distance may be, for example, a total distance by which the visual line position has moved, but is a distance indicating a radius or a diameter of a range in which the visual line position has moved with reference to an initially detected visual line position. The variation distance of the visual line position is calculated by, for example, an average value, a maximum value, or the like of the variation distance of the visual line position indicated by all of the visual line position data stored in RAM **46**. The predetermined distance is set to, for example, a distance at which it is recognized that the visual line position is gazed at. The predetermined distance is an example of a “first distance” according to the technology of the present disclosure.

(66) The visual line position display control unit **106** outputs the visual line position data including the visual line position mark **202** that is a mark indicating the visual line position on the captured image **200** to the display **72** in a case of determining that the variation distance of the visual line position is equal to or shorter than the predetermined distance. The visual line position indicated by the visual line position mark **202** is set based on, for example, an average value or the like of the visual line position indicated by all of the visual line position data stored in the RAM **46**.

(67) The display **72** displays the visual line position mark **202** indicated by the visual line position data by superimposing the visual line position mark **202** on the captured image **200**. In this way, the visual line position mark **202** corresponding to the visual line position is displayed on the display **72**.

(68) Subsequently, the visual line position display control unit **106** determines whether or not a state where the variation distance of the visual line position is equal to or shorter than the predetermined distance continues for a predetermined time or longer. The predetermined time is set to, for example, a time during which it is recognized that the visual line position is gazed at. The predetermined time is an example of a “first time” according to the technology of the present disclosure. The above-described processing of acquiring the captured image data (refer to FIG. 5), the above-described processing of acquiring the visual line data (refer to FIG. 6), and the above-described processing of outputting the visual line position data are executed again in a case where the visual line position display control unit **106** determines that a state where the variation distance of the visual line position is equal to or shorter than the predetermined distance does not continue for the predetermined time or longer. Accordingly, the visual line position data is repeatedly output to the display **72** in a case where a state where the variation distance of the visual line position is equal to or shorter than the predetermined distance does not continue for the predetermined time or longer. As described above, the visual line position data indicating the visual line position mark **202** specifying the visual line position in the captured image **200** is output to the display **72** until the predetermined time is reached. The visual line position data is an example of “data indicating a visual line position mark” according to the technology of the present disclosure. The visual line position mark **202** is an example of a “visual line position mark” according to the technology of the present disclosure.

(69) The visual line position display control unit **106** changes a form of the visual line position mark **202** with the passage of time in a case of repeatedly outputting the visual line position data to the display **72**. For example, the visual line position display control unit **106** performs processing of intensifying color of the visual line position mark **202** with the passage of time. The visual line position display control unit **106** may perform, for example, processing of intensifying a form other than the color, such as a shape, of the visual line position mark **202** with the passage of time.

(70) As shown in FIG. 8 as an example, in a case where the visual line position display control unit **106** determines that a state where the variation distance of the visual line position is equal to or shorter than the predetermined distance continues for the predetermined time or longer, the gaze position display control unit **108** detects, as a gaze position, a visual line position of a case where a state where a variation distance of the visual line position with respect to the captured image **200** is equal to or shorter than the predetermined distance continues for the predetermined time or longer. Then, the gaze position display control unit **108** outputs the gaze position data including a gaze position mark **204** that is a mark indicating the gaze position on the captured image **200** to the display **72**. As described above, the gaze position data indicating the gaze position mark **204** specifying the gaze position in the captured image **200** is output to the display **72** in a case where the predetermined time is reached. The display **72** displays the gaze position mark **204** indicated by the gaze position data by superimposing the gaze position mark **204** on the captured image **200** instead of the visual line position mark **202**.

(71) It is desirable that the gaze position mark **204** has a form different from that of the visual line position mark **202** in order to indicate that the gaze position has been detected from the visual line position. In the example shown in FIG. 8, the gaze position mark **204** is displayed in a color more intense than the visual line position mark **202**. The gaze position is an example of a “gaze position” according to the technology of the present disclosure. The gaze position data is an example of “data indicating a gaze position mark” according to the technology of the present disclosure. The gaze position mark **204** is an example of a “gaze position mark” according to the technology of the present disclosure.

(72) After outputting gaze position data to the display **72**, the gaze position display control unit **108** deletes the visual line position data stored in the RAM **46**. Further, the gaze position display control unit **108** generates gaze position data indicating a gaze position and stores the gaze position data in the RAM **46** in a case of detecting the gaze position. Then, the above-described processing from the acquisition of the captured image data to the storage of the gaze position data into RAM **46** is repeatedly executed, whereby a plurality of pieces of gaze position data are stored in the RAM **46**.

(73) As shown in FIG. **9** as an example, the target region setting unit **110** sets a target region **206** for the captured image **200** based on a plurality of gaze positions. Specifically, the target region setting unit **110** performs the following processing. That is, the target region setting unit **110** acquires all the gaze position data stored in the RAM **46** since the target region setting processing is started. Subsequently, the target region setting unit **110** determines whether or not the number of the detected gaze positions reaches a predetermined number based on the acquired gaze position data. The number of detected gaze positions corresponds to the number of the pieces of gaze position data stored in the RAM **46**. The predetermined number is set to a number with which the target region **206** can be set for the captured image **200** based on the gaze positions. In the first embodiment, for example, the predetermined number is set to two.

(74) The above-described processing from the acquisition of the captured image data to the output of the gaze position data is executed again in a case where the target region setting unit **110** determines that the number of the detected gaze positions does not reach the predetermined number. FIG. **9** shows a state where two gaze positions are detected and two gaze position marks **204** are displayed on the display **72** by repeatedly executing the processing from the acquisition of the captured image data to the output of the gaze position data.

(75) The target region setting unit **110** sets the target region **206** for the captured image **200** based on the detected gaze positions in a case of determining that the number of the detected gaze positions reaches the predetermined number. The target region setting unit **110** sets, for example, a rectangular region, which has the gaze positions indicated by the two gaze position marks **204** as opposite corners, as the target region **206** for the captured image **200**. In this way, the target region **206** is set for the captured image **200** based on the plurality of gaze positions. Then, the target region setting unit **110** outputs target region data including a target region frame **208** indicating the target region **206** on the captured image **200** to the display **72**. The display **72** displays the target region frame **208** indicated by the target region data by superimposing the target region frame **208** on the captured image **200** and the gaze position mark **204**. The target region **206** is an example of a “first region” according to the technology of the present disclosure.

(76) The target region **206** set in the above-described manner is utilized for, for example, adjustment of an exposure through automatic exposure, adjustment of an image quality through auto white balance, and/or the like. The target region setting processing may be executed from the beginning in the above-described process of setting a region in a case where the receiving unit **94** (refer to FIG. **2**) receives an instruction to delete the gaze positions. Accordingly, it is possible to set the gaze position again in a case where the gaze position is set at a position that the user does not desire.

(77) In the first embodiment, the predetermined number is set to two as an example, but the predetermined number may be set to three or more. That is, the number of the gaze positions to be detected may be three or more.

(78) Next, an action of the imaging apparatus according to the first embodiment will be described with reference to FIG. **10**. FIG. **10** shows an example of a flow of the target region setting processing according to the first embodiment.

(79) In the target region setting processing shown in FIG. **10**, first, in step ST**10**, the captured image display control unit **102** causes the image sensor **24** to image the subject. The captured image display control unit **102** acquires captured image data obtained by imaging the subject with the image sensor **24**. After the processing of step ST**10** is executed, the target region setting

processing transitions to step ST11.

(80) In step ST11, the captured image display control unit **102** outputs the captured image data acquired in step ST10 to the display **72**. Accordingly, the captured image **200** is displayed on the display **72**. After the processing of step ST11 is executed, the target region setting processing transitions to step ST12.

(81) In step ST12, the visual line detection processing unit **104** acquires visual line data obtained by being imaged by the visual line sensor **86**. After the processing of step ST12 is executed, the target region setting processing transitions to step ST13.

(82) In step ST13, the visual line detection processing unit **104** detects the visual line position with respect to the display **72** based on the visual line data acquired in step ST12. After the processing of step ST13 is executed, the target region setting processing transitions to step ST14.

(83) In step ST14, the visual line detection processing unit **104** generates the visual line position data indicating the visual line position detected in step ST13, and stores the visual line position data in the RAM **46**. After the processing of step ST14 is executed, the target region setting processing transitions to step ST15.

(84) In step ST15, the visual line position display control unit **106** acquires all the visual line position data stored in RAM **46** since the target region setting processing is started, and determines whether or not the variation distance of the visual line position is equal to or shorter than the predetermined distance based on the acquired visual line position data. In a case where the variation distance of the visual line position is not equal to or shorter than the predetermined distance in step ST15, the determination is negative and the target region setting processing transitions to step ST10. In a case where the variation distance of the visual line position is equal to or shorter than the predetermined distance in step ST15, the determination is positive and the target region setting processing transitions to step ST16.

(85) In step ST16, the visual line position display control unit **106** outputs the visual line position data including the visual line position mark **202** indicating the visual line position to the display **72**. Accordingly, the visual line position mark **202** indicated by the visual line position data is displayed on the display **72** while being superimposed on the captured image **200**. After the processing of step ST16 is executed, the target region setting processing transitions to step ST17.

(86) In step ST17, the visual line position display control unit **106** determines whether or not a state where the variation distance of the visual line position is equal to or shorter than the predetermined distance continues for the predetermined time or longer. In a case where a state where the variation distance of the visual line position is equal to or shorter than the predetermined distance does not continue for the predetermined time or longer in step ST17, the determination is negative and the target region setting processing transitions to step ST10. In a case where a state where the variation distance of the visual line position is equal to or shorter than the predetermined distance continues for the predetermined time or longer in step ST17, the determination is positive and the target region setting processing transitions to step ST18.

(87) In step ST18, the gaze position display control unit **108** detects the visual line position as the gaze position, and outputs the gaze position data including the gaze position mark **204** indicating the detected gaze position to the display **72**. Accordingly, the gaze position mark **204** indicated by the gaze position data is displayed on the display **72** while being superimposed on the captured image **200**. After the processing of step ST18 is executed, the target region setting processing transitions to step ST19.

(88) In step ST19, the gaze position display control unit **108** deletes the visual line position data stored in the RAM **46**. After the processing of step ST19 is executed, the target region setting processing transitions to step ST20.

(89) In step ST20, the gaze position display control unit **108** generates gaze position data indicating the gaze position detected in step ST18, and stores the gaze position data in the RAM **46**. After the processing of step ST20 is executed, the target region setting processing transitions to step ST21.

(90) In step ST21, the target region setting unit **110** acquires all gaze position data stored in the RAM **46** since the target region setting processing is started, and determines whether or not the number of the detected gaze positions reaches the predetermined number based on the acquired gaze position data. In a case where the number of the detected gaze positions does not reach the predetermined number in step ST21, the determination is negative and the target region setting processing transitions to step ST10. In a case where the number of the detected gaze positions reaches the predetermined number in step ST21, the determination is positive and the target region setting processing transitions to step ST22.

(91) In step ST22, the target region setting unit **110** sets the target region **206** for the captured image **200** based on the plurality of gaze positions indicated by the plurality of pieces of gaze position data stored in the RAM **46** in step ST20. After the processing of step ST22 is executed, the target region setting processing transitions to step ST23.

(92) In step ST23, the target region setting unit **110** outputs the target region data including the target region frame **208** indicating the target region **206** set in step ST22 to the display **72**. Accordingly, the target region frame **208** indicated by the target region data is displayed on the display **72** while being superimposed on the captured image **200** and the gaze position mark **204**. After the processing of step ST23 is executed, the target region setting processing ends. The above-described information processing method described as the action of the imaging apparatus is an example of an “information processing method” according to the technology of the present disclosure.

(93) As described above, in the first embodiment, the processor **42** acquires the captured image data and outputs the captured image data to the display **72**. Further, the processor **42** acquires the visual line data and detects, as a gaze position, a visual line position of a case where a state where a variation distance of the visual line position with respect to the captured image **200** indicated by the captured image data is equal to or shorter than the predetermined distance continues for the predetermined time or longer based on the visual line data. Then, the processor **42** sets the target region **206** based on the two or more gaze positions. Therefore, it is possible to set the target region **206** for the captured image **200** based on the visual line position.

(94) Further, the processor **42** sets a rectangular region, which has two gaze positions as opposite corners, as the target region **206**. Therefore, it is possible to set the target region **206** that is a rectangular region for the captured image **200** based on two visual line positions.

(95) In addition, the processor **42** outputs the visual line position data indicating the visual line position mark **202** that specifies the visual line position in the captured image **200** until the predetermined time is reached, and outputs the gaze position data indicating the gaze position mark **204** specifying the gaze position in the captured image **200** in a case where the predetermined time is reached. Accordingly, it is possible to display the gaze position mark **204** at a gazed position by gazing the captured image **200** until the predetermined time is reached.

(96) Further, the visual line position mark **202** changes in form with the passage of time. Therefore, for example, compared to a case where the form of the visual line position mark **202** is constant, it is possible to allow the user to recognize that the captured image **200** is being gazed.

(97) As shown in FIG. **11** as an example, the target region setting unit **110** may set a circular region having a line connecting two gaze positions as a diameter as the target region **206** for the captured image **200** in a case of setting the target region **206** based on the gaze position.

(98) Further, the region set based on two or more gaze positions may have any shape.

Second Embodiment

(99) Next, a second embodiment will be described.

(100) FIG. **12** shows the second embodiment. In the second embodiment, the first embodiment is modified as follows. Hereinafter, the second embodiment will be described based on an example in which four gaze positions are detected.

(101) In the second embodiment, in a case where it is necessary to distinguish and describe a

plurality of gaze position marks **204** corresponding to the plurality of gaze positions, the plurality of gaze position marks **204** are referred to as a gaze position mark **204A**, a gaze position mark **204B**, a gaze position mark **204C**, and a gaze position mark **204D**. In the example shown in FIG. **12**, each gaze position corresponding to each gaze position mark **204** is detected in an order of the gaze position mark **204A**, the gaze position mark **204B**, the gaze position mark **204C**, and the gaze position mark **204D**.

(102) In the second embodiment, for example, the predetermined number for setting the target region **206** based on the gaze positions is set to four.

(103) The target region setting unit **110** acquires all the gaze position data stored in the RAM **46** since the target region setting processing is started, and determines whether or not the number of the detected gaze positions reaches the predetermined number based on the acquired gaze position data.

(104) The target region setting unit **110** sets the target region **206** for the captured image **200** in a case of determining that the number of the detected gaze positions reaches the predetermined number. For example, the target region setting unit **110** sets, as the target region **206** for the captured image **200**, a closed region defined by connecting four gaze positions in an order in which the four gaze positions indicated by the four gaze position marks **204** have been detected. Then, the target region setting unit **110** outputs the target region data including the target region frame **208** that is a mark indicating the target region **206** on the captured image **200** to the display **72**. The display **72** displays the target region frame **208** indicated by the target region data by superimposing the target region frame **208** on the captured image **200** and the gaze position mark **204**.

(105) In the second embodiment, the processor **42** sets, as the target region **206** for the captured image **200**, a closed region defined by connecting the four gaze positions in the order in which the four gaze positions have been detected. Therefore, it is possible to set the target region **206** that is a closed region for the captured image **200** based on four visual line position.

(106) In the second embodiment, the target region **206** may be set based on three gaze positions, or may be set based on five or more gaze positions.

Third Embodiment

(107) Next, a third embodiment will be described.

(108) FIG. **13** shows the third embodiment. In the third embodiment, the first embodiment is modified as follows. In the third embodiment, the gaze position is referred to as a gaze position **210**. Hereinafter, the third embodiment will be described based on an example in which five gaze positions **210** are detected.

(109) In the third embodiment, in a case where it is necessary to distinguish and describe a plurality of gaze positions **210**, the plurality of gaze positions **210** are referred to as a gaze position **210A**, a gaze position **210B**, a gaze position **210C**, a gaze position **210D**, and a gaze position **210E**. In the example shown in FIG. **13**, the gaze positions **210** are detected in an order of the gaze position **210A**, the gaze position **210B**, the gaze position **210C**, the gaze position **210D**, and the gaze position **210E**.

(110) The target region setting unit **110** acquires all the gaze position data stored in the RAM **46** since the target region setting processing is started, and determines whether the target region **206** (refer to FIG. **14**) can be set based on the plurality of gaze positions **210** indicated by the acquired gaze position data. For example, as in the example shown in FIG. **13**, the target region setting unit **110** determines that the target region **206** can be set based on the plurality of gaze positions **210** in a case where the target region **206** can be set based on the gaze position **210B**, the gaze position **210C**, the gaze position **210D**, and an intersection position **214** as a position at which a line **212A** connecting two gaze positions **210A** and **210B** and a line **212B** connecting two gaze positions **210D** and **210E** intersect is set as the intersection position **214** in a case where three or more gaze positions **210** are detected. The target region setting unit **110** sets a position at which the lines **212A**

and **212B** connecting two gaze positions **210** intersect with each other as the intersection position **214** in a case of determining that the target region **206** can be set based on the plurality of gaze positions **210**.

(111) Subsequently, as shown in FIG. **14** as an example, the target region setting unit **110** sets the target region **206** based on the gaze position **210B**, the gaze position **210C**, the gaze position **210D**, and the intersection position **214** that are present at positions capable of defining the target region **206**. That is, a closed region defined by connecting the gaze position **210B**, the gaze position **210C**, the gaze position **210D**, and the intersection position **214** is set as the target region **206** for the captured image **200**. As described above, in a case where the target region **206** having four vertices is set, at least 4+1 gaze positions **210** are used.

(112) Then, the target region setting unit **110** outputs the target region data including the target region frame **208** that is a mark indicating the target region **206** on the captured image **200** to the display **72**. Further, the target region setting unit **110** outputs the gaze position data including the gaze position marks **204** that are marks indicating the gaze positions **210** defining the target region **206** on the captured image **200** and intersection position data including an intersection position mark **216** that is a mark indicating the intersection position **214** on the captured image **200** to the display **72**. The display **72** superimposes and displays the gaze position mark **204**, the intersection position mark **216**, and the target region frame **208** on the captured image **200** based on the gaze position data, the intersection position data, and the target region data. Accordingly, instead of the plurality of gaze position marks **204** displayed on the display **72** corresponding to the five gaze positions **210**, the gaze position marks **204**, the intersection position mark **216**, and the target region frame **208** that define the target region **206** are displayed.

(113) In the third embodiment, in a case where three or more gaze positions **210** are detected, the processor **42** sets a position at which lines **212A** and **221B** connecting two gaze positions **210** intersect with each other as the intersection position **214**, and sets the target region **206** based on the gaze positions **210** and the intersection position **214** present at positions capable of defining the target region **206**. Therefore, even in a case where the lines **212A** and **212B** connecting two gaze positions **210** are generated in a process of setting the target region **206**, it is possible to set the target region **206** that is a closed region.

(114) In the third embodiment, the target region **206** is set based on five gaze positions **210**, but may be set based on four gaze positions **210**. In this case, a triangular target region **206** is set. As described above, in a case where the target region **206** having three vertices is set, at least 3+1 gaze positions **210** are used. Further, the target region **206** may have a shape having n vertices. n is a natural number. When the target region **206** having n vertices is set, at least 4+1 gaze positions **210** are used.

Fourth Embodiment

(115) Next, a fourth embodiment will be described.

(116) FIGS. **15** and **16** show the fourth embodiment. In the fourth embodiment, the first embodiment is modified as follows.

(117) As shown in FIG. **15** as an example, in the fourth embodiment, a CPU operates as a track display control unit **112**. The track display control unit **112** causes the display **72** to display a track line **218** corresponding to the visual line position. Specifically, the track display control unit **112** performs the following processing. That is, the track display control unit **112** acquires all the visual line position data stored in the RAM **46** since the target region setting processing is started. Subsequently, the visual line position display control unit **106** detects a track defined by connecting a plurality of visual line positions in an order in which the plurality of visual line positions are detected, based on a plurality of pieces of acquired visual line position data. Then, the visual line position display control unit **106** outputs track data including the track line **218** indicating the track to the display **72**. The display **72** displays the track line **218** indicated by the track data while superimposing the track line **218** on the captured image **200**. In this way, the track line **218**

corresponding to the visual line positions is displayed on the display **72**. The track is an example of a “track” according to the technology of the present disclosure.

(118) The track display control unit **112** determines whether or not a state where the track is detected continues for a predetermined time or longer. The predetermined time may be, for example, a time from start to end of reception of an instruction from the user through the receiving unit **94** (refer to FIG. **2**). Further, the predetermined time may be a time until a predetermined time (for example, a time in which a track for setting a region can be secured) elapses from the start of the reception of the instruction from the user, or may be a time from when the user starts to direct a visual line to the display **72** until the user removes the visual line from the display **72**. The predetermined time is an example of a “second time” according to the technology of the present disclosure.

(119) The track display control unit **112** executes again the above-described processing of acquiring the captured image data (refer to FIG. **5**), the above-described processing of acquiring the visual line data (refer to FIG. **6**), and the above-described processing of outputting the track data in a case of determining that a state where the track is detected does not continue for the predetermined time or longer. Accordingly, in a case where a state where the track is detected does not continue for the predetermined time or longer, the track data is repeatedly output to the display **72**. As described above, until the predetermined time is reached, the track along which the visual line position has varied is detected, and the track data including the track line **218** that is a line indicating the detected track is output to the display **72**. In the fourth embodiment, the captured image data is an example of “second image data” according to the technology of the present disclosure. The captured image **200** indicated by the captured image data is an example of a “second image” according to the technology of the present disclosure.

(120) As shown in FIG. **16** as an example, the target region setting unit **110** sets a target region **206** for the captured image **200** based on the detected track. Specifically, the target region setting unit **110** performs the following processing. That is, the target region setting unit **110** sets a closed region surrounding the track indicated by the track line **218** as the target region **206** for the captured image **200** in a case where the track display control unit **112** determines that a state the track is detected continues for the predetermined time or longer. In this case, the target region setting unit **110** may set a closed region circumscribing the track as the target region **206** for the captured image **200**, or may set a closed region surrounding the track away from the track as the target region **206** for the captured image **200**. In this way, the target region **206** is set for the captured image **200** based on the track. The target region **206** is an example of a “second region” according to the technology of the present disclosure.

(121) Then, the target region setting unit **110** outputs the target region data including the target region frame **208** that is the mark indicating the target region **206** on the captured image **200** to the display **72**. The display **72** displays the target region frame **208** indicated by the target region data by superimposing the target region frame **208** on the captured image **200** and the track line **218**.

(122) Next, an action of the imaging apparatus according to the fourth embodiment will be described with reference to FIG. **17**. FIG. **17** shows an example of a flow of the target region setting processing according to the fourth embodiment.

(123) In the target region setting processing shown in FIG. **17**, the processing from step ST**30** to step ST**34** is the same as the processing from step ST**10** to step ST**14** in the target region setting processing according to the first embodiment (refer to FIG. **10**). After the processing of step ST**34** is executed, The target region setting processing shown in FIG. **17** transitions to step ST**35**.

(124) In step ST**35**, the track display control unit **112** acquires all the visual line position data stored in the RAM **46** since the target region setting processing is started, and detects the track defined by connecting the plurality of visual line positions in the order in which the plurality of visual line positions are detected, based on the plurality of pieces of acquired visual line position data. Then, the visual line position display control unit **106** outputs the track data including the

track line **218** that is a mark indicating the track to the display **72**. Accordingly, the track line **218** indicated by the track data is displayed on the display **72** while being superimposed on the captured image **200**.

(125) In step **ST36**, the track display control unit **112** determines whether or not a state where the track is detected continues for the predetermined time or longer. In a case where a state where the track is detected does not continue for the predetermined time or longer in step **ST36**, the determination is negative and the target region setting processing transitions to step **ST30**. In a case where a state where the track is detected continues for the predetermined time or longer in step **ST36**, the determination is positive, and the target region setting processing transitions to step **ST37**.

(126) In step **ST37**, the target region setting unit **110** sets a closed region surrounding the track detected in step **ST35** as the target region **206** for the captured image **200**. After the processing of step **ST37** is executed, the target region setting processing transitions to step **ST38**.

(127) In step **ST38**, the target region setting unit **110** outputs the target region data including the target region frame **208** indicating the target region **206** set in step **ST37** to the display **72**.

Accordingly, the target region frame **208** indicated by the target region data is displayed on the display **72** while being superimposed on the captured image **200**. After the processing of step **ST38** is executed, the target region setting processing ends. The above-described information processing method described as the action of the imaging apparatus is an example of an “information processing method” according to the technology of the present disclosure. As described above, in the fourth embodiment, the processor **42** acquires the captured image data and outputs the captured image data to the display **72**. Further, the processor **42** acquires the visual line data, and detects a track along which the visual line position has varied within the predetermined time with respect to the captured image **200** indicated by the captured image data based on the visual line data. Then, the processor **42** sets the target region **206** based on the track. Therefore, it is possible to set the target region **206** for the captured image **200** based on the track along which the visual line position has varied.

(128) Further, the processor **42** sets a closed region surrounding the track as the target region **206** for the captured image **200**. Therefore, it is possible to set the target region **206**, which is a closed region surrounding the track, based on the track along which the visual line position has varied.

(129) As shown in FIG. **18** as an example, the target region setting unit **110** may set a closed region (for example, a rectangular region) inscribed in the track indicated by the track line **218** as the target region **206** for the captured image **200**.

(130) Further, as shown in FIG. **19** as an example, in a case where the track indicated by the track line **218** forms a closed region, the target region setting unit **110** may set the closed region formed by the track as the target region **206** for the captured image **200**. That is, a region surrounded by the track may be set as the target region **206** for the captured image **200**.

Fifth Embodiment

(131) Next, a fifth embodiment will be described.

(132) FIG. **20** shows the fifth embodiment. In the fifth embodiment, the fourth embodiment is modified as follows.

(133) The track display control unit **112** determines whether or not a blink occurs, and outputs track data generated based on a determination result to the display **72**. Specifically, the track display control unit **112** performs the following processing. That is, the track display control unit **112** determines whether or not a blink occurs within a time during which the track is being detected, based on the visual line position data stored in the RAM **46**. The visual line position data stored in RAM **46** includes visual line position data acquired in a case where the eye is blinking (that is, in a case where the eye **302** is closed) in addition to visual line position data acquired in a case where the eye is not blinking (that is, in a case where the eye **302** is open).

(134) In a case where the visual line position data acquired in a case where the eye is blinking (that

is, in a case where the eye **302** is closed) is detected, the track display control unit **112** determines that a blink occurs. For example, the visual line position data acquired in a case where the eye is blinking (that is, in a case where the eye **302** is closed) may include error information indicating that the visual line position is not detected. In a case where the visual line position data acquired from the RAM includes visual line position data including error information, the track display control unit **112** determines that a blink occurs.

(135) In a case of determining that a blink occurs, the track display control unit **112** detects a track by executing linear interpolation that connects a first visual line position before the occurrence of the blink and a second visual line position after the occurrence of the blink based on the visual line position data acquired before and after the visual line position data in a case of blinking. That is, the track display control unit **112** detects the track based on the line connecting the first visual line position before the occurrence of the blink and the second visual line position after the occurrence of the blink. Then, the track display control unit **112** outputs track data indicating the track to the display **72**. In this way, the track data generated based on the result of determining whether or not a blink occurs is output to the display **72**. The display **72** displays the track line **218** indicated by the track data while superimposing the track line **218** on the captured image **200**.

(136) As shown in FIG. **20** as an example, a first visual line position mark **202A** indicating the first visual line position, a second visual line position mark **202B** indicating the second visual line position, and an interpolation line **218A** indicating a line connecting the first visual line position and the second visual line position may be displayed while being superimposed on the captured image **200**. Further, in a case where the first visual line position and the second visual line position are separated from each other to such an extent that linear interpolation cannot be performed, the track display control unit **112** may cause the display **72** to display text or the like indicating an error.

(137) Next, an action of the imaging apparatus according to the fifth embodiment will be described with reference to FIG. **21**. FIG. **21** shows an example of a flow of the target region setting processing according to the fifth embodiment.

(138) In the target region setting processing shown in FIG. **21**, processing from step **ST40** to step **ST41** is added to the target region setting processing according to the fourth embodiment (refer to FIG. **17**). The processing from step **ST30** to step **ST34** is the same as the target region setting processing according to the fourth embodiment. After the processing of step **ST34** is executed, the target region setting processing shown in FIG. **17** transitions to step **ST40**.

(139) In step **ST40**, the track display control unit **112** determines whether or not a blink occurs within the time during which the track is being detected, based on the visual line position data acquired in step **ST34**. In a case where a blink does not occur within the time during which the track is being detected in step **ST40**, the determination is negative and the target region setting processing transitions to step **ST35**. In a case where a blink occurs within a time during which the track is being detected in step **ST40**, the determination is positive and the target region setting processing transitions to step **ST41**.

(140) In a case of determining in step **ST41** that a blink occurs, the track display control unit **112** detects a track by performing linear interpolation between the first visual line position before the occurrence of the blink and the second visual line position after the occurrence of the blink based on visual line position data acquired before and after the visual line position data in the case of blinking. The processing from step **ST35** to step **ST38** is the same as the target region setting processing according to the fourth embodiment.

(141) As described above, in the fifth embodiment, the processor **42** determines whether or not a blink occurs within the time during which the track is being detected based on the visual line position data obtained from the visual line data. In a case of determining that a blink occurs, the processor **42** detects a track based on the line connecting the first visual line position before the occurrence of the blink and the second visual line position after the occurrence of the blink.

Accordingly, even in a case where a blink occurs, it is possible to set the target region **206** based on the track.

Sixth Embodiment

(142) Next, a sixth embodiment will be described.

(143) FIG. **22** to FIG. **25** show the sixth embodiment. In the sixth embodiment, the first embodiment is modified as follows.

(144) As shown in FIG. **22** as an example, the captured image display control unit **102** causes the display **72** to display a division grid **222** that divides the captured image **200** into a plurality of regions **220**. Specifically, the captured image display control unit **102** performs the following processing. That is, the captured image display control unit **102** causes the image sensor **24** to image the subject (not shown) through the image sensor control circuit **26** by outputting the imaging control signal to the image sensor control circuit **26**. The captured image display control unit **102** acquires captured image data obtained by imaging the subject with the image sensor **24**.

(145) Subsequently, the captured image display control unit **102** divides the captured image **200** into the plurality of regions **220**. In this case, the captured image display control unit **102** may divide the captured image **200** into the plurality of regions **220** according to an instruction from the user received by the receiving unit **94** (refer to FIG. **2**), or may divide the captured image **200** into the plurality of regions **220** based on a result of performing image processing on the captured image data. Further, the captured image display control unit **102** may divide the captured image **200** into a plurality of regions **220** based on information relating to a depth of field or the like obtained from the captured image data. The plurality of regions **220** are an example of a “plurality of regions” according to the technology of the present disclosure.

(146) Then, the captured image display control unit **102** generates region division data indicating the division grid **222** that divides the captured image **200** into the plurality of regions **220**, and outputs the captured image data and the region division data to the display **72**. The display **72** displays the captured image **200** indicated by the captured image data. Accordingly, the captured image **200** in which the subject appears as an image is displayed on the display **72**. Further, the display **72** displays the division grid **222** indicated by the region division data by superimposing the division grid **222** on the captured image **200**. In this way, the division grid **222** that divides the captured image **200** into the plurality of regions **220** is displayed on the display **72**.

(147) As shown in FIG. **23** as an example, the visual line position display control unit **106** causes the display **72** to display the visual line position mark **202** corresponding to the visual line position by outputting the visual line position data to the display **72**, as in the first embodiment. The visual line position display control unit **106** changes the form of the visual line position mark **202** with the passage of time in a case of repeatedly outputting the visual line position data to the display **72**. For example, the visual line position display control unit **106** intensifies the color of the visual line position mark **202** with the passage of time.

(148) As shown in FIG. **24** as an example, the gaze position display control unit **108** detects the visual line position as the gaze position as in the first embodiment. Then, the gaze position display control unit **108** outputs the gaze position data including the gaze position mark **204** that is the mark indicating the gaze position on the captured image **200** to the display **72**, thereby causing the display **72** to display the gaze position mark **204** by superimposing the gaze position mark **204** on the captured image **200**. Then, the plurality of gaze positions are detected by repeatedly executing the processing from the acquisition of the captured image data to the detection of the gaze position.

(149) FIG. **25** shows a state where two gaze positions are detected and two gaze position marks **204** are displayed on the display **72** by repeatedly executing the processing from the acquisition of the captured image data to the detection of the gaze position.

(150) The target region setting unit **110** sets the target region **206** from the plurality of regions **220** based on the plurality of gaze positions. In this case, for example, the target region setting unit **110** extracts, from the plurality of regions **220**, a region falling within a rectangular region having two

gaze positions indicated by the two gaze position marks **204** as opposite corners, and sets the extracted region as the target region **206** for the captured image **200**. Then, the target region setting unit **110** outputs the target region data including the target region frame **208**, which is a frame indicating the target region **206**, on the captured image **200** to the display **72**. The display **72** displays the target region frame **208** indicated by the target region data by superimposing the target region frame **208** on the captured image **200**.

(151) Note that, in the sixth embodiment, a region is set from the plurality of regions **220** included in the captured image **200** based on two gaze positions, but a region may be set from the plurality of regions **220** included in the captured image **200** based on three or more gaze positions.

(152) Next, an action of the imaging apparatus according to the sixth embodiment will be described with reference to FIG. **26**. FIG. **26** shows an example of a flow of the target region setting processing according to the sixth embodiment.

(153) In the target region setting processing shown in FIG. **26**, a process of step ST**50** is added to the target region setting processing (refer to FIG. **10**) according to the first embodiment. After the processing of step ST**10** is executed, the target region setting processing shown in FIG. **27** transitions to step ST**50**.

(154) In step ST**50**, the captured image display control unit **102** generates region division data indicating the division grid **222** that divides the captured image **200** into the plurality of regions **220**. After the processing of step ST**50** is executed, the target region setting processing transitions to step ST**11**.

(155) In step ST**11**, the captured image display control unit **102** outputs the captured image data acquired in step ST**10** and the region division data generated in step ST**50** to the display **72**. Accordingly, the captured image **200** indicated by the captured image data is displayed on the display **72**. Further, the division grid **222** indicated by the region division data is displayed on the display **72** while being superimposed on the captured image **200**. After the processing of step ST**11** is executed, the target region setting processing transitions to step ST**12**.

(156) The processing from step ST**12** to step ST**21** is the same as the target region setting processing according to the first embodiment.

(157) In step ST**22**, the target region setting unit **110** sets the target region **206** from the plurality of regions **220** based on the plurality of gaze positions indicated by the plurality of pieces of gaze position data stored in the RAM **46** in step ST**20**. After the processing of step ST**22** is executed, the target region setting processing transitions to step ST**23**.

(158) The processing of step ST**23** is the same as the target region setting processing according to the first embodiment.

(159) As described above, in the fifth embodiment, the processor **42** sets the target region **206** from the plurality of regions **220** included in the captured image **200** based on two or more gaze positions. Therefore, it is possible to set the target region **206** on the captured image **200** based on the visual line position.

(160) Note that the target region setting unit **110** may extract, from the plurality of regions **220**, a region that falls within a circular region having a line connecting the two gaze positions as a diameter, and set the extracted region **220** as the target region **206**. Further, the target region setting unit **110** may extract, from the plurality of regions **220**, a region that falls within a closed region defined by connecting the plurality of gaze positions in the order in which the plurality of gaze positions are detected, and set the extracted region **220** as the target region **206**.

Seventh Embodiment

(161) Next, a seventh embodiment will be described.

(162) FIGS. **27** and **28** show the seventh embodiment. In the seventh embodiment, the fourth embodiment is modified as follows.

(163) As shown in FIG. **27** as an example, the captured image display control unit **102** causes the display **72** to display a division grid **222** that divides the captured image **200** into the plurality of

regions **220**, as in the sixth embodiment. Similarly to the fourth embodiment, the track display control unit **112** outputs the track data including the track line **218**, which is a line indicating the track, to the display **72**, thereby displaying the track line **218** indicated by the track data by superimposing the track line **218** on the captured image **200**.

(164) As shown in FIG. **28** as an example, the target region setting unit **110** sets the target region **206** from the plurality of regions **220** based on the track indicated by the track line **218**. In this case, for example, the target region setting unit **110** extracts the region **220** overlapping with the track from the plurality of regions **220** and sets the extracted region **220** as the target region **206** for the captured image **200**.

(165) Then, the target region setting unit **110** outputs the target region data including the target region frame **208**, which is a frame indicating the target region **206**, on the captured image **200** to the display **72**. The display **72** displays the target region frame **208** indicated by the target region data by superimposing the target region frame **208** on the captured image **200**.

(166) Next, an action of the imaging apparatus according to the seventh embodiment will be described with reference to FIG. **29**. FIG. **29** shows an example of a flow of the target region setting processing according to the seventh embodiment.

(167) In the target region setting processing shown in FIG. **29**, processing of step ST**60** is added to the target region setting processing (refer to FIG. **17**) according to the fourth embodiment. After the processing of step ST**30** is executed, the target region setting processing shown in FIG. **29** transitions to step ST**60**.

(168) In step ST**60**, the captured image display control unit **102** generates the region division data indicating the division grid **222** that divides the captured image **200** into the plurality of regions **220**. After the processing of step ST**60** is executed, the target region setting processing transitions to step ST**31**.

(169) In step ST**31**, the captured image display control unit **102** outputs the captured image data acquired in step ST**30** and the region division data generated in step ST**60** to the display **72**. Accordingly, the captured image **200** indicated by the captured image data is displayed on the display **72**. Further, the division grid **222** indicated by the region division data is displayed on the display **72** while being superimposed on the captured image **200**. After the processing of step ST**31** is executed, the target region setting processing transitions to step ST**32**. The processing from step ST**32** to step ST**36** is the same as the target region setting processing according to the fourth embodiment.

(170) In step ST**37**, the target region setting unit **110** sets the target region **206** from the plurality of regions **220** based on the track detected in step ST**35**. After the processing of step ST**37** is executed, the target region setting processing transitions to step ST**38**.

(171) The processing of step ST**38** is the same as the target region setting processing according to the fourth embodiment.

(172) As described above, in the seventh embodiment, the processor **42** sets the target region **206** from the plurality of regions **220** included in the captured image **200** based on the track. Therefore, it is possible to set the target region **206** for the captured image **200** based on the track detected in accordance with the variation of the visual line position.

(173) Note that the target region setting unit **110** may extract a region surrounding the track from the plurality of regions **220** and set the extracted region **220** as the target region **206** for the captured image **200**. Further, the target region setting unit **110** may extract a region surrounded by the track from the plurality of regions **220** and set the extracted region **220** as the target region **206** for the captured image **200**.

Eighth Embodiment

(174) Next, an eighth embodiment will be described.

(175) FIG. **30** shows the eighth embodiment. In the eighth embodiment, the first embodiment is modified as follows.

(176) As shown in FIG. 30 as an example, the processor 42 operates as an autofocus processing unit 114. In a case where a target subject 224 is included as an image in the target region 206 in the captured image 200, the autofocus processing unit 114 performs control to bring the target subject 224 within the depth of field.

(177) Specifically, the autofocus processing unit 114 performs the following processing. That is, first, the autofocus processing unit 114 executes target region setting processing. For example, the target region setting processing according to any one of the first to seventh embodiments described above is applied to the target region setting processing. Accordingly, the target region 206 is set for the captured image 200, and the captured image 200 and the target region frame 208 are displayed on the display 72.

(178) Subsequently, the autofocus processing unit 114 determines whether or not the target subject 224 is included as an image in the target region 206 in the captured image 200.

(179) In this case, for example, the autofocus processing unit 114 determines whether or not the target subject 224 is included as an image in the target region 206, based on the captured image data acquired in the target region setting processing by using an image processing technology such as pattern matching. The target subject 224 refers to a subject to be brought within the depth of field as described below. The target subject 224 is an example of a “target subject” according to the technology of the present disclosure.

(180) In the example shown in FIG. 30, the target subject 224 is a person's face. The target subject 224 may be a face of an animal or a subject other than a face, in addition to a face of a person. Further, in a case of determining that the target subject 224 is included as an image in the target region 206, a detection frame 226 indicating a frame surrounding the target subject 224 may be displayed on the display 72 while being superimposed on the captured image 200.

(181) Then, in a case of determining that the target subject 224 is included as an image in the target region 206, the autofocus processing unit 114 performs the control to bring the target subject 224 within the depth of field. That is, the autofocus processing unit 114 outputs a lens control signal to the lens control circuit 62 to control the first actuator 64 through the lens control circuit 62, thereby adjusting the position of the focus lens 56 to a position where the target subject 224 is brought within the depth of field. In this way, the control to bring the target subject 224 within the depth of field is performed.

(182) Next, an action of the imaging apparatus according to the eighth embodiment will be described with reference to FIG. 31. FIG. 31 shows an example of a flow of the autofocus processing according to the eighth embodiment.

(183) In the autofocus processing shown in FIG. 31, first, in step ST70, the autofocus processing unit 114 executes the target region setting processing. Accordingly, the target region 206 is set for the captured image 200, and the captured image 200 and the target region frame 208 are displayed on the display 72.

(184) After the processing of step ST70 is executed, the autofocus processing transitions to step ST71.

(185) In step ST71, the autofocus processing unit 114 determines whether or not the target subject 224 is included as an image in the target region 206 based on the target region data acquired by the target region setting processing executed in step ST70. In a case where the target subject 224 is not included as an image in the target region 206 in step ST71, the determination is negative and the autofocus processing ends. In a case where the target subject 224 is included as an image in the target region 206 in step ST71, the determination is positive and the autofocus processing transitions to step ST72.

(186) In step ST72, the autofocus processing unit 114 performs the control to bring the target subject 224 within the depth of field. Accordingly, the target subject 224 is brought within the depth of field. After the processing of step ST72 is executed, the autofocus processing ends.

(187) As described above, in the eighth embodiment, in a case where the target subject 224 is

included as an image in the target region **206**, the processor **42** performs control to bring the target subject **224** within the depth of field. Accordingly, for example, in the autofocus processing, the target subject **224** can be brought within the depth of field based on the visual line position.

(188) Although the first to eighth embodiments have been described above, the above-described embodiments and modification examples can be combined with each other as long as no contradiction arises. In a case where the above-described embodiments and modification examples are combined, in a case where there are a plurality of overlapping steps, priorities may be given to the plurality of steps depending on various conditions or the like.

(189) In the above-described embodiments, a digital camera is exemplified as the imaging apparatus **10**, but the technology of the present disclosure is not limited thereto, and for example, a terminal device functioning as the imaging apparatus **10** (for example, a tablet terminal, a smart device, or a wearable terminal) may be used. Further, the digital camera serving as the imaging apparatus **10** may be any of various digital cameras such as a compact digital camera, a mirrorless single-lens reflex camera, and a digital single-lens reflex camera. The digital camera may be a lens-interchangeable digital camera or a lens-fixed digital camera. Further, the technology of the present disclosure may be applied to various information processing apparatuses other than the imaging apparatus **10** (for example, a cell observation apparatus, an ophthalmic observation apparatus, a surgical microscope, or the like).

(190) In the above-described embodiments, the visual line position of one eye is detected, but visual line positions of both eyes may be detected.

(191) Further, the processor **42** has been exemplified in each of the above-described embodiments, but at least one other CPU, at least one GPU, and/or at least one TPU may be used instead of the processor **42** or together with the processor **42**.

(192) In the above-described embodiment, although an embodiment example in which the program **100** is stored in the NVM **44** has been described, the technology of the present disclosure is not limited thereto. For example, the program **100** may be stored in a portable non-transitory computer-readable storage medium such as an SSD or a USB memory (hereinafter, simply referred to as a “non-transitory storage medium”). The program **100** stored in the non-transitory storage medium is installed on the computer **20** of the imaging apparatus **10**, and the processor **42** executes processing according to the program **100**.

(193) Further, the program **100** may be stored in a storage device such as another computer or a server device connected to the imaging apparatus **10** through a network, and the program **100** may be downloaded in response to a request from the imaging apparatus **10** and installed in the computer **20**.

(194) Further, it is not necessary to store the entire program **100** in a storage device such as another computer or a server device connected to the imaging apparatus **10** or in the NVM **44**, and a part of the program **100** may be stored.

(195) Further, although the computer **20** is incorporated in the imaging apparatus **10**, the technology of the present disclosure is not limited thereto, and for example, the computer **20** may be provided outside the imaging apparatus **10**.

(196) Further, although the computer **20** including the processor **42**, the NVM **44**, and the RAM **46** is exemplified in each of the above-described embodiments, the technology of the present disclosure is not limited thereto, and a device including at least one of an ASIC, an FPGA, or a PLD may be applied instead of the computer **20**. Further, a combination of a hardware configuration and a software configuration may be used instead of the computer **20**.

(197) Further, the following various processors can be used as hardware resources for executing various kinds of processing described in each of the above-described embodiments. Examples of the processors include a CPU that is a general-use processor executing software, that is, a program to function as a hardware resource executing various kinds of processing. Further, examples of the processors include a dedicated electrical circuit such as an FPGA, a PLD, or an ASIC that is a

processor having a circuit configuration dedicatedly designed to execute a specific type of processing. A memory is incorporated in or connected to any processor, and any processor executes various kinds of processing by using the memory.

(198) The hardware resource executing various kinds of processing may be configured by one of these various processors, or may be configured by a combination of two or more processors of the same type or different types (for example, a combination of a plurality of FPGAs or a combination of a CPU and an FPGA). Further, the hardware resource executing various kinds of processing may be one processor.

(199) As an example in which the hardware resource is configured by one processor, first, there is a form in which one processor is configured by a combination of one or more CPUs and software, and this processor functions as the hardware resource executing various kinds of processing. Second, as represented by SoC or the like, there is a form in which a processor that realizes the functions of the entire system including a plurality of hardware resources executing various kinds of processing with one IC chip is used. As described above, various kinds of processing are realized by using one or more of the various processors described above as hardware resources.

(200) As a hardware structure of these various processors, more specifically, an electrical circuit in which circuit elements such as semiconductor elements are combined can be used. Further, the above-described visual line detection processing is merely an example. Therefore, needless to say, unnecessary steps may be deleted, new steps may be added, or the processing order may be changed without departing from the scope of the present invention.

(201) Above described contents and shown contents are detailed description for parts according to the technology of the present disclosure and are merely one example of the technology of the present disclosure. For example, the above description regarding the configuration, function, action, and effect is a description regarding an example of the configuration, function, action, and effect of the parts according to the technology of the present disclosure. Accordingly, needless to say, unnecessary parts may be deleted, new elements may be added, or replacement may be made with respect to the above described contents and the above shown contents without departing from the scope of the technology of the present disclosure. Further, in order to avoid complication and facilitate understanding of the parts according to the technology of the present disclosure, description related to common technical knowledge or the like that does not need to be particularly described in order to enable implementation of the technology of the present disclosure is omitted in the above described contents and the above shown contents.

(202) In the present specification, “A and/or B” has the same meaning as “at least one of A or B”. That is, “A and/or B” means that only A may be used, only B may be used, or a combination of A and B may be used. Further, in the present specification, in a case where three or more matters are expressed by “and/or” in combination, the same concept as “A and/or B” is applied.

(203) All documents, patent applications, and technical standards described in the present specification are herein incorporated by reference to the same extent as if each individual publication, patent application, or technical standard was specifically and individually indicated to be incorporated by reference.

Claims

1. An information processing apparatus comprising: a processor, wherein the processor is configured to: acquire first image data; output the first image data to a display; acquire visual line data; detect, as a gaze position, a visual line position of a case where a state where a variation distance of the visual line position with respect to a first image indicated by the first image data is equal to or shorter than a first distance continues for a first time or longer based on the visual line data; and set a first region based on two or more gaze positions.

2. The information processing apparatus according to claim 1, wherein the processor is configured

to set, as the first region, a rectangular region having two gaze positions as opposite corners, a circular region having a line connecting the two gaze positions as a diameter, or a closed region defined by connecting a plurality of gaze positions in an order in which the plurality of gaze positions are detected.

3. The information processing apparatus according to claim 1, wherein the processor is configured to: in a case where three or more gaze positions are detected, set a position at which lines connecting two gaze positions intersect with each other as an intersection position; and set the first region based on the gaze positions and the intersection position.

4. The information processing apparatus according to claim 3, wherein, in a case where the first region having n vertices is set, at least $n+1$ gaze positions are used.

5. The information processing apparatus according to claim 1, wherein the processor is configured to: output data indicating a visual line position mark that specifies the visual line position in the first image until the first time is reached; and output data indicating a gaze position mark that specifies the gaze position in the first image in a case where the first time is reached.

6. The information processing apparatus according to claim 5, wherein the visual line position mark changes in form with passage of time.

7. The information processing apparatus according to claim 1, wherein the processor is configured to set the first region from a plurality of regions included in the first image based on the two or more gaze positions.

8. The information processing apparatus according to claim 1, wherein the information processing apparatus is an imaging apparatus, and the processor is configured to perform, in a case where a target subject is included as an image in the first region, control to bring the target subject in a depth of field.

9. An information processing method comprising: acquiring first image data; outputting the first image data to a display; acquiring visual line data; detecting, as a gaze position, a visual line position of a case where a state where a variation distance of the visual line position with respect to a first image indicated by the first image data is equal to or shorter than a first distance continues for a first time or longer based on the visual line data; and setting a first region based on two or more gaze positions.

10. A non-transitory computer-readable storage medium storing a program for causing a computer to execute a process comprising: acquiring first image data; outputting the first image data to a display; acquiring visual line data; detecting, as a gaze position, a visual line position of a case where a state where a variation distance of the visual line position with respect to a first image indicated by the first image data is equal to or shorter than a first distance continues for a first time or longer based on the visual line data; and setting a first region based on two or more gaze positions.
